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Your review report

Manuscript

A model to improve the forecasting accuracy of Interrupted Time Series

Feedback for the author(s)

Review file(s)

gru_review.pdf

Comments to the author(s)

The manuscript would benefit from a clearer and more rigorous conceptual framing of interrupted time series forecasting. At present, the motivation for combining ARIMA, GARCH, and GRU models is only briefly stated and needs stronger theoretical grounding. The mathematical presentation of the individual models (ARIMA, GARCH, GRU) should be improved, with equations written in a consistent notation and with clearer explanations of underlying assumptions. Likewise, the hybrid modelling strategy and the proposed weighting mechanism require fuller justification, both conceptually and mathematically, so that readers can understand how each component contributes to forecasting performance.

The methodology section requires additional detail on data preprocessing, stationarity checks, handling of outliers, and the exact train–test split used in the empirical study. Hyperparameters for GRU and tuning choices for ARIMA and GARCH should be explicitly stated. Furthermore, the evaluation of model performance would be stronger if supported by statistical significance tests (e.g., Diebold–Mariano test) to determine whether forecast differences between models are meaningful rather than due to sampling variability.

The presentation of results should also be revised for clarity and conciseness. Certain sections are overly descriptive, while others lack sufficient explanation. A more streamlined narrative would help readers better understand how the findings support the manuscript's main contributions.

The literature review could be better structured to move from foundational concepts to recent developments in hybrid time series forecasting.

Standardising mathematical notation throughout the manuscript will improve readability. Figures should be refined for higher resolution and consistent formatting. Finally, it would be helpful to include an appendix containing model-tuning details (such as GRU architecture, learning rate, number of epochs, and ARIMA/GARCH selection criteria), so that the study is more reproducible.

Confidential feedback for the Editor

Your recommendation	• ***Revise***
Is the experimental procedure or study design appropriate to answer the research question (including the use of appropriate controls)? Are the conclusions supported by the evidence presented?	• ***Yes***
Please rate the novelty of the research on a scale of 1-5 (1=not at all novel and 5=extremely novel)?	• ***3***
Is the presentation of the work clear, with regards to language and grammar?	• ***Needs some language corrections before being published***
Does the title and abstract of this paper clearly reflect its content? And are the keywords representative of the research?	• ***Yes***
Does the introduction present the purpose of the research investigation and is the purpose supported by the pertinent literature?	• ***Yes***
Are the methods described sufficiently to allow the research study to be repeated by other parties? Is the software data code provided sufficient to be replicated?	• ***No***
Comments	***please refer to pdf document uploaded in the author section***
If applicable to the paper? Is the use of statistics and treatment of uncertainties appropriate?	• ***Not relevant to this manuscript***

Are the images in this manuscript free from improper manipulation?	• ***Yes***
Are the tables and figures well designed and necessary?	• ***Yes***
Are the references provided appropriate and up to date?	• ***Yes***

Confidential comments to the Editor